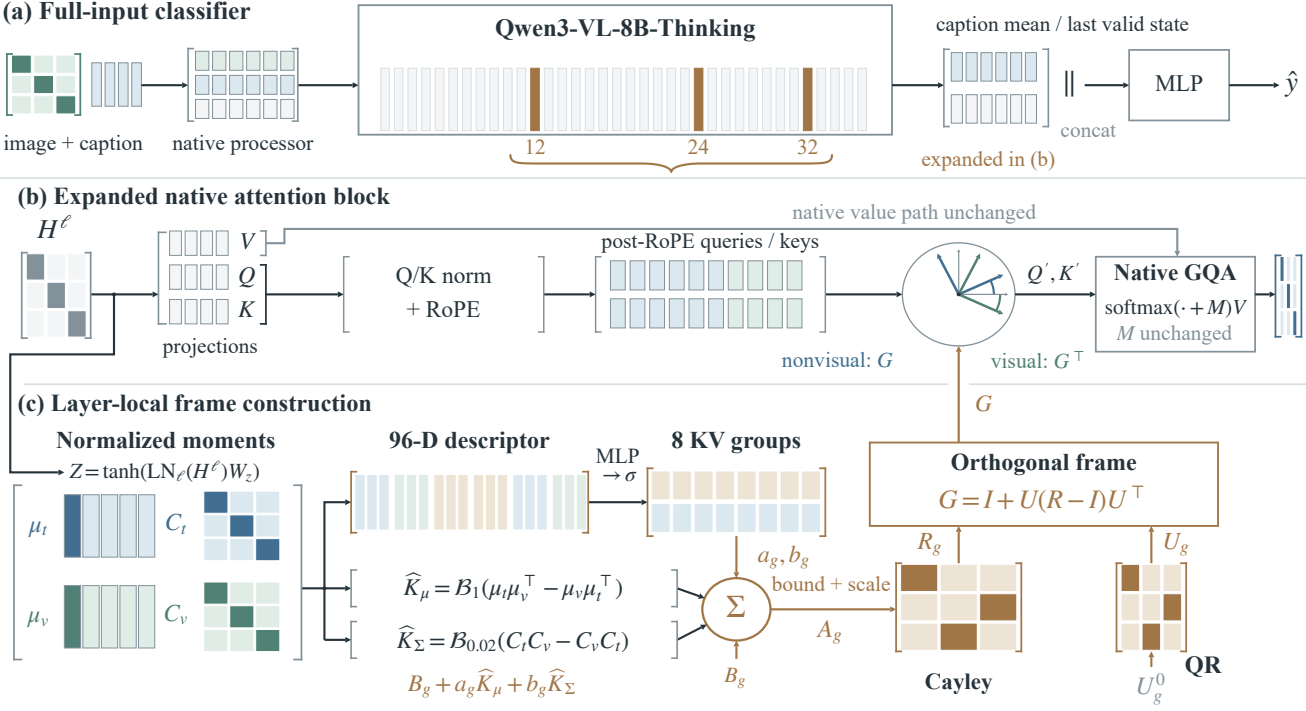

(a) Full-input classifier
image + caption
native processor
Qwen3-VL-8B-Thinking
12
24
32
caption mean / last valid state
concat
MLP
$\hat{y}$
expanded in (b)
(b) Expanded native attention block
H^ℓ
V
Q
K
projections
Q/K norm
+ RoPE
post-RoPE queries / keys
native value path unchanged
Q', K'
Native GQA
$\mathrm{softmax}(\cdot + M)V$
M unchanged
nonvisual: G
visual: $G^\top$
G
(c) Layer-local frame construction
Normalized moments
$Z = \tanh(\mathrm{LN}_\ell(H^\ell)W_z)$
μ_t
C_t
μ_v
C_v
96-D descriptor
MLP
$\to \sigma$
8 KV groups
$\widehat{K}_\mu = \mathcal{B}_1(\mu_t\mu_v^\top - \mu_v\mu_t^\top)$
$\widehat{K}_\Sigma = \mathcal{B}_{0.02}(C_tC_v - C_vC_t)$
$B_g + a_g\widehat{K}_\mu + b_g\widehat{K}_\Sigma$
a_g, b_g
bound + scale
Σ
A_g
B_g
Orthogonal frame
$G = I + U(R - I)U^\top$
R_g
U_g
Cayley
QR
U_g^0